

SEDG Greenhouse Gas Emissions (GHG) Calculator

User Guide

For Small and Medium Enterprises (SMEs) in
Malaysia

July 2025

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1. Background and Purpose

This **Greenhouse Gas (GHG) Emissions Calculator** is developed to help small and medium-sized enterprises (SMEs) in Malaysia understand and track their greenhouse gas emissions. It is based on international standards and adjusted to suit Malaysia's local context.

The calculator was created to make it easier for businesses to measure Scope 1 and Scope 2 emissions. These are emissions that come directly from the company's own operations (e.g. fuel use) and indirectly through the company's energy (e.g., electricity) use.

Tracking these emissions is the first step for any company that wants to reduce its impact on climate change, support sustainability, and prepare for reporting requirements that may come.

2. Introduction to the GHG Emissions Calculator

This tool is designed to be simple and user-friendly, especially for companies that may not have employees with expertise in sustainability or environmental science. It follows widely accepted international standards to ensure consistency and credibility.

2.1 Standards Used

The calculator is built following international and local guidelines:

International Standards:

- **GHG Protocol Corporate Accounting and Reporting Standard**
Developed by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD), GHG Protocol is the most used global standard for calculating GHG emissions. For this calculator, the latest published emission factors from the GHG Protocol were used from the "Emission Factors from Cross Sector Tools Version 2.0 (March 2024)". The latest Global Warming Potentials (GWPs) were taken from the GHG Protocol's "IPCC Global Warming Potential Values Version 2.0 (August 2024)",
- **Intergovernmental Panel on Climate Change (IPCC)**
The raw emission factors for fuel use and Global Warming Potentials

(GWPs) used by GHG Protocol come from the IPCC's 6th Assessment Report (AR6). IPCC data is the most used global standard by companies operating in countries that have not published their own national emission factors (e.g., Malaysia).

Malaysia-Specific Guidelines:

- **Malaysia's Biennial National Communications to the UNFCCC**

Malaysia in its national GHG inventory reported to the United Nations Framework Convention on Climate Change (UNFCCC) also uses the GHG protocol methodology and emission factors & GWPs from the IPCC.

- **Suruhanjaya Tenaga (Energy Commission)**

Suruhanjaya Tenaga publishes data on the greenhouse gas (GHG) intensities of grid electricity for Peninsular Malaysia, Sabah and Sarawak respectively. For this calculator, the latest available data from the "Grid Emission Factor (GEF) in Malaysia (2017-2022)" was used, with subsequent years using the latest 2022 emission factor.

2.2 Setting Your Business Boundary

When calculating your emissions, you need to decide which parts of your business to include. There are three ways to set the boundary for your business:

1. Operational Control
2. Financial Control
3. Equity Share

We recommend using the "Operational Control" approach as suggested by the GHG Protocol. That means:

*Only account for the Scope 1 & 2 emissions from operations or facilities where your company has operational control like offices, warehouses, factories, equipment or company vehicles. For example, if you have a joint venture where you have 50% equity share but 0% operational control, you do **NOT** have to account for any of the emissions. However, if you have 50% equity share but 100% operational control, you have to account for **ALL** the emissions.*

For more information regarding setting boundaries for GHG emissions accounting, please refer to the "GHG Protocol Corporate Accounting and Reporting Standard – Chapters 3 and 4".

2.3 Calculation Approach

Greenhouse gas (GHG) emissions are grouped into three categories called “scopes”. For most SMEs in Malaysia, the most urgent and manageable are Scope 1 and Scope 2:

- Scope 1 (Direct emissions) – from fuel you burn in your business operations (e.g. diesel for trucks, natural gas for boilers)
- Scope 2 (Indirect emissions) – from the electricity you purchase and use

Scope 1	Scope 2
▪ Stationary Combustion: Burning fuel in equipment like boilers or generators ▪ Mobile Combustion: Fuel used in company-owned vehicles ▪ Fugitive Emission: Leaks from equipment like air-conditioners or refrigerators (e.g. refrigerant gases)	▪ Purchased of electricity from the grid

2.3.1 Scope 1 – Direct Emissions

1. Stationary Combustion

Scope 1 stationary combustion refers to emissions from burning fuels like diesel or natural gas in equipment such as boilers, generators, or furnaces owned or controlled by the company.

Estimated emissions are calculated by multiplying the relevant activity data (e.g. fuel consumption in litre) with the appropriate emission factor.

Formula:

$$CO_2e \text{ emissions} = \sum Activity \text{ Data} \times Emission \text{ Factor} \times GWP$$

where,

- Activity data refers to the measured amount of a specific activity that generates emissions. For example, litres of diesel fuel used in a boiler
- Emission factor refers to the average amount of a specific greenhouse gas emitted per unit of activity. Among these, CO₂ typically accounts for the largest portion of emissions, while CH₄ and N₂O have much smaller contributions
- GWP is the measure of how much heat a greenhouse gas traps in the atmosphere over a specific time (usually 100 years), relative to carbon dioxide

Major Greenhouse Gases	GWP values for 100-year time horizon, AR6
Carbon dioxide (CO ₂)	1.0
Methane – non-fossil (CH ₄)	27.0
Methane – fossil (CH ₄)	29.8
Nitrous oxide (N ₂ O)	273.0

Example:

Suppose a bill value is 10,000 m³ of natural gas. The emission factors are as follows:

1884 gCO₂/m³

0.168 gCH₄/m³

0.003 gN₂O/m³

Calculating Emissions:

$$CO_2 \text{ emissions} = 10,000 \text{ m}^3 \times 1,884 \text{ gCO}_2/\text{m}^3 = 18,840 \text{ kgCO}_2$$

$$CH_4 \text{ emissions} = 10,000 \text{ m}^3 \times 0.168 \text{ gCH}_4/\text{m}^3 = 1.68 \text{ kgCH}_4$$

$$N_2O \text{ emissions} = 10,000 \text{ m}^3 \times 0.003 \text{ gN}_2\text{O}/\text{m}^3 = 0.03 \text{ kgN}_2\text{O}$$

Converting the emissions to CO₂e by multiplying GWP:

$$CO_2e \text{ emissions} = 18,840 \text{ kgCO}_2 \times 1 = 18,840 \text{ kgCO}_2e$$

$$CO_2e \text{ emissions} = 1.68 \text{ kgCH}_4 \times 29.8 = 50.064 \text{ kgCO}_2e$$

$$CO_2e \text{ emissions} = 0.03 \text{ kgN}_2\text{O} \times 273 = 8.19 \text{ kgCO}_2e$$

Total emissions:

$$CO_2e \text{ emissions} = (18,840 + 50.064 + 8.19) \text{ kgCO}_2e = 18,898.254 \text{ kgCO}_2e$$

2. Mobile combustion

Mobile combustion refers to fuel burned by vehicles owned or controlled by the company. This includes road vehicles like cars, vans, lorries, and trucks.

Exactly as with stationary combustion, estimated emissions are calculated by multiplying the relevant activity data (e.g. fuel consumption in litre) with the appropriate emission factor.

Formula:

$$CO_2e \text{ emissions} = \sum Activity \text{ Data} \times Emission \text{ Factor} \times GWP$$

where,

- Activity data refers to the measured amount of fuel consumed, recorded in volume (e.g. litres) to standardise across different modes of transport.
- As with stationary combustion, emission factors include CO₂, CH₄, and N₂O emitted for every unit of fuel consumption.
- GWP is the measure of how much heat a greenhouse gas traps in the atmosphere over a specific time (usually 100 years), relative to carbon dioxide

4. Fugitive emissions from refrigerants (excluded)

Fugitive emissions come from the leakage of refrigerants, which are gases used in refrigeration and air conditioning equipment such as chillers, freezers, cold rooms, and air conditioners. For most SMEs, this is not a significant source of GHG emissions e.g., if operating only small air-conditioning units. As a result, this source of emissions has been **excluded** from the SEDG GHG Emissions Calculator.

For SMEs operating in industries with heavy refrigeration activities however e.g., operating refrigerated trucks or warehouses for food logistics, it may be necessary to carry out the full inventory approach as described in the GHG Protocol's - "Calculating HFC and PFC Emissions from the Manufacturing, Installation, Operation and Disposal of Refrigeration & Airconditioning Equipment (Version 1.0)".

5. Physical and chemical process emissions (excluded)

In addition, physical and chemical processes such as cement, steel or waste processing also generate GHG emissions.

In most cases, cement and steel plants as well as landfills are operated by large companies and not SMEs. As a result, this source of emissions has been **excluded** from the SEDG GHG Emissions Calculator.

For SMEs operating in these industries, it may be necessary to account for process emissions as described by the guidance documents in the GHG Protocol's "Sector-Specific Tools".

2.3.2 Scope 2 – Indirect Emissions from Purchased Electricity

Estimated emissions are calculated by multiplying the electricity purchased from the grid with the Grid Emission Factor (GEF).

$$CO_2e \text{ emissions} = \text{Electricity purchased} \times GEF$$

where,

- The electricity purchased is typically measured in kilowatt-hours (kWh) or megawatt-hours (MWh) and can be obtained from the monthly electricity bills charged by the electricity providers
- In Malaysia, purchased electricity comes from three main electricity grids:
 - Peninsular Malaysia supplied by Tenaga Nasional Bhd (TNB)
Note Kulim Hi-Tech Park is a special case and has its own provider – N.U.R Power Sdn. Bhd. with its own GEF.
 - Sabah supplied by Sabah Electricity Sdn. Bhd. (SESB)
 - Sarawak supplied by Sarawak Energy Bhd.
- GEF refers to the average amount of carbon dioxide equivalent (tonnes CO₂e) emitted per Megawatt-hour (MWh) of electricity generated and supplied by each respective grid
- Different grids have different GEFs because they generate the electricity using different methods. For example, Peninsula Malaysia has a higher share of fossil fuel power plants (e.g. coal and natural gas) and so it has the highest GEF, whereas Sarawak has predominantly hydropower (renewable energy), and so it has the lowest GEF
- The GEF of each grid changes every year, as the generation mix (i.e., the % of different energy sources) of the grid changes

2.4 Emission Factors

The calculator comes with pre-filled specific emission factors, including for:

- Fuel combustion commonly used in Malaysia (diesel, petrol, LPG, natural gas)
- Grid electricity consumption from Malaysia's three main grids and Kulim Hi-Tech Park

2.4.1 Scope 1 Direct Emissions

For this calculator, the latest published emission factors from the GHG Protocol were used from the "Emission Factors from Cross Sector Tools Version 2.0 (March 2024)". The latest Global Warming Potentials (GWPs) were taken from the GHG Protocol's "IPCC Global Warming Potential Values Version 2.0 (August 2024)". The raw emission factors for fuel use and Global Warming Potentials (GWPs) used by GHG Protocol come from the IPCC's 6th Assessment Report (AR6). Where necessary, volume-to-weight conversions were made using average fuel densities from sources like the EPA.

a. Stationary Combustion

Fuel Type	Emission Factor	Unit
Coal	1.82	mt CO ₂ e/mt
LPG	0.0632	mt CO ₂ e/GJ
Natural Gas	0.0562	mt CO ₂ e/GJ
Residual Fuel Oil (Bunker fuel oil)	0.00304	mt CO ₂ e/litres
Gasoline (Petrol)	0.00232	mt CO ₂ e/litres
Diesel	0.00295	mt CO ₂ e/litres
Biodiesel B100	0.00067	mt CO ₂ e/mt
Diesel B20	0.00236	mt CO ₂ e/litres
Diesel B10	0.00265	mt CO ₂ e/litres
Diesel B7	0.00274	mt CO ₂ e/litres
Diesel B5	0.00280	mt CO ₂ e/litres

b. Mobile Combustion

Fuel Type	Emission Factor	Unit
Gasoline (Petrol)	0.002288	mt CO ₂ e/litres
Diesel	0.002909	mt CO ₂ e/litres
Diesel B7	0.002705	mt CO ₂ e/litres
Diesel B10	0.002618	mt CO ₂ e/litres

Diesel B20	0.002327	mt CO ₂ e/litres
Biodiesel B100	0.000001	mt CO ₂ e/litres
LPG	0.001473	mt CO ₂ e/litres
CNG	0.001885	mt CO ₂ e/litres

2.4.2 Scope 2 Indirect Emissions

For this calculator, the latest available emission factors from Suruhanjaya Tenaga published in the "Grid Emission Factor (GEF) in Malaysia (2017-2022)" were used, with subsequent years using the latest 2022 emission factor. For Kulim Hi-Tech Park, the emission factor published by N.U.R Power for 2023 "NUR Power's Scope 1 & 2 Emissions in 2023" is used for all years.

a. Purchased Electricity

Year	Location	Emission Factor	Unit
2022	Peninsular Malaysia - Tenaga Nasional Bhd	0.774	mt CO ₂ e/MWh
	Sabah - Sabah Electricity Sdn Bhd	0.525	mt CO ₂ e/MWh
	Sarawak - Sarawak Energy Bhd	0.199	mt CO ₂ e/MWh
	Kulim Hi-Tech Park – N.U.R Power	0.540	mt CO ₂ e/MWh
2023	Peninsular Malaysia - Tenaga Nasional Bhd	0.774	mt CO ₂ e/MWh
	Sabah - Sabah Electricity Sdn Bhd	0.525	mt CO ₂ e/MWh
	Sarawak - Sarawak Energy Bhd	0.199	mt CO ₂ e/MWh
	Kulim Hi-Tech Park – N.U.R Power	0.540	mt CO ₂ e/MWh
2024	Peninsular Malaysia - Tenaga Nasional Bhd	0.774	mt CO ₂ e/MWh
	Sabah - Sabah Electricity Sdn Bhd	0.525	mt CO ₂ e/MWh
	Sarawak - Sarawak Energy Bhd	0.199	mt CO ₂ e/MWh
	Kulim Hi-Tech Park – N.U.R Power	0.540	mt CO ₂ e/MWh

3. Guide to Using the GHG Calculator

CMM's SEDG GHG Emissions Calculator has a simple web interface for you to calculate your emissions. At the homepage, simply click on "Start Calculating" to get to the data input section.



Step 1: Company Information

The screenshot displays the "Step 1: Company Information" form within the SEDG GHG Emissions Calculator interface. The top navigation bar remains the same. Below the navigation bar, a breadcrumb trail shows the progression: "Overview > Scope 1 > Scope 2 > Intensity > Result Summary". The main heading is "Welcome to the Simplified GHG Emissions Calculator". A paragraph explains that the calculator is designed to help Malaysian SMEs easily calculate their Scope 1 and 2 emissions by simply entering their activity data as specified. Another paragraph states that to begin, basic information about the company is requested to better understand the GHG emission profiles across different sectors in Malaysia, and that this information will be used solely by Capital Markets Malaysia for internal tracking and analysis, with the final report remaining private and accessible only to the user. The form itself contains three dropdown menus: "Industry" with a help icon, "Company Size" with a help icon, and "Company Revenue" with a help icon. Each dropdown menu has a placeholder text "-- Select Type --", "-- Select Range --", and "-- Select Range --" respectively. A blue "Next" button is located at the bottom left of the form. The footer of the page contains the copyright notice: "Copyright © 2025 Capital Markets Malaysia. All Rights Reserved."

You are first required to input some information about your company via drop-down menus – your industry, company size in terms of number of employees and revenue range. No other information about your company is required, i.e., this data is effectively anonymous and will be used solely by CMM for internal tracking and analysis at industry level.

Then click “Next”.

Note that throughout the calculator there are ⓘ signs that you can hover over for short explanations about each data input.

Step 2: Scope 1 Data

From the drop-down menus, you may select your fuels for both stationary combustion and mobile combustion. To add more than one fuel, click on “+ Add more”. You may add up to five different fuels for each category. Then select the unit for which you wish to input the data for each fuel. There are different units available for each fuel, please be careful to input the correct unit as recorded in your data sheets or invoices.

Finally, you may input the annual consumption data for up to three years as available – 2022, 2023 and 2024. If you do not have data for a particular year, please leave that field **blank**.

Activity data may be aggregated from purchasing invoices and receipts as shown in the examples below:

Stationary combustion (Natural Gas):



BIL GAS DAN INVOIS

Keterangan Caj	GJ	RM / GJ	Amaun (RM)
PENGUNAAN GAS	147700 Sm3		
	5633 GJ		
Pelarasan	0 GJ		
Caj Penggunaan	5633	x	
Caj Gas	5633	x	53.78
Caj Penghantaran	5633	x	1.128
Caj Kapasiti (Pengalihan)	15841	x	1.715/GJ
Caj "Authorised Capacity Overrun"	0	x	2.058/GJ
Jumlah Bil Semasa			

If you have a boiler burning natural gas, you will likely have invoices from Gas Malaysia

In the invoice, they already tell you exactly how many Gigajoules of gas you have purchased.

Mobile combustion (Petrol/Diesel):



Statement Period : 01/02/2020 - 29/02/2020

Date/Time	Station	Receipt no.	Product	Quantity (L)	Unit price (RM)	Amount
26/02/2020, 07:28am	PETRONAS Grand Saga 2	RYB0670202002250101	PRIMAX 95	24.04	2.08	RM50.00
23/02/2020, 21:46pm	PETRONAS Grand Saga 2	RYB0670202002230082	PRIMAX 95	28.85	2.08	RM60.00
22/02/2020, 20:31pm	PETRONAS Lot 353 Cheras	RYW0510202002220035	PRIMAX 95	28.85	2.08	RM60.00
19/02/2020, 10:15am	PETRONAS Grand Saga 2	RYB0670202002190021	PRIMAX 95	29.13	2.06	RM60.00
16/02/2020, 11:34am	PETRONAS Batu 8 Jalan Ipoh	RYW0042202002160012	PRIMAX 95	29.13	2.06	RM60.00
13/02/2020, 23:39pm	PETRONAS Lot 279 Hulu Klang	RYB0594202002130133	PRIMAX 95	29.41	2.04	RM60.00
11/02/2020, 09:46am	PETRONAS Prima Saujana 1	RYB0769202002110014	PRIMAX 95	29.41	2.04	RM60.00
09/02/2020, 14:11pm	PETRONAS Sg Bakap Layby	RYP1306202002090008	PRIMAX 95	12.26	2.04	RM25.00
08/02/2020, 21:44pm	PETRONAS Batu Feringgi	RYP1067202002080015	PRIMAX 95	26.96	2.04	RM55.00
08/02/2020, 04:25am	PETRONAS Rawang RSA	RYB0437202002070136	PRIMAX 95	24.51	2.04	RM50.00
06/02/2020, 20:12pm	PETRONAS Grand Saga 2	RYB0670202002060062	PRIMAX 95	24.04	2.08	RM50.00
06/02/2020, 05:47am	PETRONAS Grand Saga 2	RYB0670202002050095	PRIMAX 95	9.62	2.08	RM20.00
05/02/2020, 08:37am	PETRONAS Grand Saga 2	RYB0670202002050010	PRIMAX 95	9.62	2.08	RM20.00
03/02/2020, 20:07pm	PETRONAS Jalan Sungai Tua	RYB0962202002030041	PRIMAX 95	9.62	2.08	RM20.00
TOTAL						RM650.00

If you have company vehicles running on petrol/diesel, you likely have a fuel card (e.g., Setel) where you account for it via statements

You would need to add up each "top up" quantity to get the total for the month.

Once the Scope 1 data has been fully inputted, click "Next".

Step 3: Scope 2 Data

The screenshot shows the 'Calculate Scope 2' section of the SEDG Simplified ESG Disclosure Guide. The page has a blue header with the CM2 logo and navigation links: Home, Methodology, User Guide, About, Start Calculating, EN, BM, CN. Below the header is a breadcrumb trail: Overview > Scope 1 > Scope 2 > Intensity > Result Summary. The main heading is 'Calculate Scope 2'. Below it, a paragraph explains that Scope 2 emissions are indirect GHG emissions from purchased energy. Another paragraph states that annual data for 2022, 2023, and 2024 is preferred. The form itself is titled 'Purchased Electricity' and includes a table with columns for Region, Unit, 2022, 2023, and 2024. The Region column has a dropdown menu with '-- Select Region --'. The Unit column has a dropdown menu with 'kWh'. The 2022, 2023, and 2024 columns have input fields, each with 'kWh' pre-filled. There is a '+ Add more' button below the table. Below the table, there is a section titled 'Businesses with Solar Installations' with two bullet points explaining how to handle solar energy. At the bottom of the form are 'Previous' and 'Next' buttons. The footer of the page says 'Copyright © 2025 Capital Markets Malaysia. All Rights Reserved.'

Region	Unit	2022	2023	2024
-- Select Region --	kWh	kWh	kWh	kWh

+ Add more

Businesses with Solar Installations:

- If you have purchased the solar outright under the Net Energy Metering (NEM) / Net Off-set Virtual Aggregation (NOVA) programme, you own the green attributes and should therefore deduct all the generated solar electricity sent back to the energy provider from the total grid consumption.
- If your solar installation is under a "Feed in Tariff" scheme, you do not retain the green attributes and therefore cannot offset your electricity consumption with the solar energy generated.

Previous Next

From the drop-down menu, you may select the region/grid from which you purchase your electricity. To add more than one, e.g., if you have factories in multiple regions, click on "+ Add more". You may add a row for all three regions in Malaysia as well as Kulim Hi-Tech Park which is a special case if necessary. Then select the unit for which you wish to input the data for each region. You may input the data in either kWh or MWh. Please be careful to input the correct unit as recorded in your data sheets or invoices. Finally, you may input the annual consumption data for up to three years as available – 2022, 2023 and 2024. If you do not have data for a particular year, please leave that field **blank**.

Activity data may be aggregated from purchasing invoices and receipts as shown in the examples below:

Peninsula Malaysia (TNB):



BIL ELEKTRIK ANDA

Jumlah Perlu Dibayar : RM 2,025,459.00 Tarikh Bil : 01.08.2021

	Amaun	Bayar Sebelum
Tunggakan	RM 0.00	Terima Kasih
Caj Semasa	RM 2,025,458.98	
Penggenapan	RM 0.02	
Jumlah Bil	RM 2,025,459.00	31.08.2021
Bil Terdahulu (01.07.2021)	RM 2,080,387.65	Bayaran Akhir (13.07.2021)
Jenis Bacaan	: Bacaan Sebenar	
Tempoh Bil	: 01.07.2021 - 31.07.2021 (31 Hari)	
Tarif	: E2:Perindustrian	
Blok Tarif (kWh/kW)	Kegunaan (kWh/kW)	Kadar(RM)
	3,442,772.00	0.3550
Kegunaan Puncak	2,404,486.00	0.2190
Kegunaan Luar Puncak	9,727.00	37.0000
Kehendak Maksima		
Jumlah		2,108,665.49

	Amaun(RM)
Kegunaan Puncak	1,222,164.06
Kegunaan Luar Puncak	826,582.43
Kehendak Maksima	359,899.00
Jumlah	2,108,665.49

	Amaun(RM)
Kegunaan kWh	5,847,258.00
Kegunaan Puncak	1,748,766.49
Kehendak Maksima	359,899.00
Rebat ICPT (RM0.02/kWh)	-116,945.16
Kegunaan Bulan Semasa	1,991,720.33
KWTBB (1.6%)	33,738.65
Caj Semasa	2,025,458.98

Muatan Tertinggi Dicatat 12,000.00 kW

No. Meter	Faktor Meter	Bacaan Meter	Kegunaan	Unit
M 821700419	1.00000	0.00	3,442,772.00	kWh P
M 821700419	1.00000	0.00	2,404,486.00	kWh O
M 821700419	1.00000	0.00	2,196,856.00	kVAh
M 821700419	1.00000	0.00	9,727.00	kW P
M 821700419	1.00000	0.00	9,495.00	kW O

Bil : LPC

Untuk maklumat bil dan bayaran terdahulu, sila layari: <http://www.mytnb.com.my> atau hubungi Hotline TNB 1 300 88 5454

Untuk gangguan bekalan atau kerosakan lampu jalan TNB sila hubungi melalui telefon/SMS: 15454

Untuk pertanyaan, sila hubungi:

TNB SHAH ALAM
PSN DAMAI SEK 11
49100 SHAH ALAM
SELANGOR
Tel : 03-55102020
Fax : 03-55103543

Surcaj 1% dikenakan ke atas bil bulanan untuk tempoh 6 bulan (1 Jul - 31 Dis 2021)

Bayaran melalui cek sah setelah penjelasan cek oleh bank

Rebat ICPT RM0.02/kWh diberikan bagi penggunaan kWh bulanan untuk tempoh 6 bulan (1 Jul - 31 Dis 2021)

Total consumption for the month is 5,847,258 kWh.

You would need to add up all 12 monthly statements

Peninsula Malaysia (TNB with NEM NOVA):



BIL ELEKTRIK ANDA

Penerangan	Penggunaan	Kadar (RM)	Amaun (RM)
Puncak (kWh)	131,049.00	0.3550	46,522.40
Luar Puncak (kWh)	117,409.00	0.2190	25,712.57
Kehendak Maksima (kW)	820.00	37.0000	30,340.00
Jumlah	248,458.00		102,574.97

Lebihan Tenaga yang Dijana Anda

Blok (kWh)	Penjanaan (kWh)	Kadar (RM)	Jumlah (RM)
Energy Rate <= HIAQ	15,521	0.1054	-3,079.37
Energy Rate > HIAQ	0		0.00
Jumlah			-3,079.37

Keterangan	Tanpa ST	Dengan ST	Jumlah
Jumlah Penggunaan Anda (248,458 kWh)	RM 72,234.97	0.00	72,234.97
Kehendak Maksima	RM 30,340.00	0.00	30,340.00
ICPT (RM0.16/kWh)	RM 39,753.28	0.00	39,753.28
Caj Penggunaan Bulan Semasa	RM 142,328.25	0.00	142,328.25
Kumpulan Wang Tenaga Boleh Baharu (1.6%)	RM 1,641.20		1,641.20
Lebihan Tenaga yang Dijana	RM -3,079.37		-3,079.37
Nett Offset	RM 0.00		0.00
Caj Semasa	RM 140,895.08		140,895.08

Saluran Pembayaran

myTNB

PERBANKAN INTERNET

EPAY (Petronas, KK Mart, Caltex)

Perlu Bantuan?

1-300-88-5454
Pertanyaan akaun dan bil

15454
Gangguan bekalan elektrik di rumah dan lampu jalan

tnbcareline@tnb.com.my

e.g., Total electricity consumption for the month is 248,458 kWh. But 15,521 kWh is your own solar generation.

So total grid electricity consumption:
248,458 kWh – 15,521 kWh = **232,937 kWh**

If you have purchased the solar system outright under NEM NOVA, you own the green attributes and so as described in the figure above, should deduct

all the generated solar electricity that is sent back to TNB "Penjanaan" from the total grid consumption "Jumlah".

If your NEM NOVA solar installation is under a Zero Capex model, please check in the contract if you own the green attributes. If you do, you may also do the same.

If your solar installation is under a "Feed in Tariff", you do not own any of the green attributes and cannot deduct any solar generation from your electricity consumption.

Once the Scope 2 data has been fully inputted, click "Next".

Step 4: Intensity (Optional)

The screenshot shows the 'Calculate Emissions Intensity' section of the SEDG Simplified ESG Disclosure Guide. The interface includes a navigation bar with links to Home, Methodology, User Guide, About, and Start Calculating. A breadcrumb trail indicates the current step: Overview > Scope 1 > Scope 2 > Intensity > Result Summary. The main heading is 'Calculate Emissions Intensity'. Below this, a paragraph explains that emissions intensity is an additional metric calculated by dividing absolute Scope 1 and 2 emissions by a business performance metric like production output, sales, or revenue. It notes that this is useful for customers who need the emission intensity of products purchased from them to measure their own Scope 3 emissions. A note states: 'Please provide annual revenue across multiple years that you have entered from the previous sections. If the data is unavailable for certain years, you may leave those fields blank.' The form contains a section titled 'Revenue (Optional if relevant)' with a table for inputting annual revenue for 2022, 2023, and 2024. The table has columns for 'Unit', '2022', '2023', and '2024'. The 'Unit' column is set to 'Malaysian Ringgit (RM)'. The '2022' column has a value of 'RM'. The '2023' and '2024' columns are empty. Below the table are 'Previous' and 'Next' buttons. The footer of the page reads 'Copyright © 2025 Capital Markets Malaysia. All Rights Reserved.'

Unit	2022	2023	2024
Annual Revenue	RM		

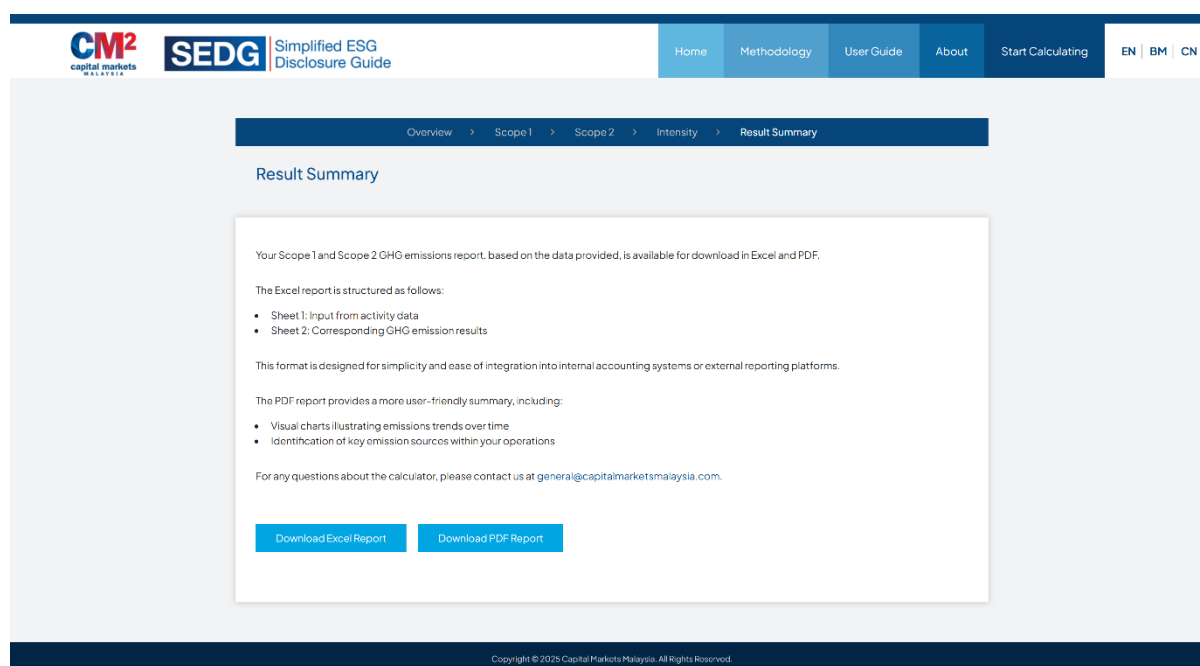
Note: Calculating the Scope 1 and 2 GHG Intensity is a completely optional step, and it involves dividing the emissions with an organisation specific metric e.g., revenue, production, etc. This is the metric that would be most useful for your customer, as they eventually need the emissions per tonne of products purchased from you to measure their own Scope 3 emissions. It also allows companies that are still growing and expanding production to still show progress over time from GHG reduction initiatives despite absolute emissions increasing.

For simplicity, this calculator only allows you to calculate your emission intensity per unit revenue and not per tonne of production. You may input the revenue of your company for any years for which you have input consumption data.

If you do not wish to calculate the emission intensity, please leave all fields blank.

Click “Next”.

Step 5: Download Results



You have completed your Scope 1 and 2 calculations!

To access your results, simply click on the “Download Excel Report” and/or “Download PDF Report” buttons.

In the Excel report, you will have your absolute annual Scope 1 & Scope 2 emissions, optional emission intensity, the detailed breakdown into Stationary Combustion, Mobile Combustion, etc. as well as the individual emission sources. The report is structured with your emission results at the top and your data inputs at the bottom for reference. The Excel report is more suited to back-end data collection and enables you to easily copy numbers into your own accounting sheets.

In the PDF report, you will have the same emission results but in a more visually pleasing format that includes charts as well for easy analysis of your biggest

sources of emissions and how they have developed over time. Data inputs are not included in the PDF report.

Thank you for using Capital Market Malaysia's SEDG GHG Emissions Calculator for SMEs. Please do not hesitate to contact us at general@capitalmarketsmalaysia.com for any questions or suggestions that you may have.